Chin-Ta (Kiida) Lai Ph.D.

Email: ctlai@ntu.edu.tw / u93132@gmail.com

EDUCATION

University of California San Diego, US (UCSD)

Ph.D. Structural Engineering (2019 - 2024)
Thesis: Dynamic Modeling of the NHERI UCSD 6-DOF Large High-Performance Outdoor Shake Table Under Bare and Loaded Table Conditions
Advisor: Professor Joel P. Conte

M.S. Mechanical and Aerospace Engineering (2019 - 2021), GPA: 3.90/4.0

National Taiwan University, Taiwan (NTU)

M.S. Structural Engineering (2014 - 2016), GPA:3.96/4.0, rank: 1/43
Thesis: System Control Strategy on Floor Response Reproduction of High-rise Buildings Subjected to Earthquakes
Advisor: Professor Keh-Chyuan Tsai, Professor Pei-Ching Chen

B.S. Civil Engineering (2010 - 2014), GPA: 3.36/4.0

Coursera

Machine Learning Foundations

EXPERIENCES

Department of Civil Engineering, National Taiwan University (NTU), Taiwan

Assistant Professor (02/2025 -)

- Course:

  2025 Spring: Physical Model Design Laboratory (CIE1011)
  Seminar in Structural Mechanics (CIE7071)

  2025 Fall: Seminar in Structural Mechanics (CIE7071)
  Structural Engineering and Fluid Mechanics Lab (CIE3005)

  2026 Spring Introduction to Robotics: Math and Practice (CIE5159)
  Physical Model Design Laboratory (CIE1011)

  2026 Fall Design of Steel Structure (CIE4001)
  Structural Engineering and Fluid Mechanics Lab (CIE3005)

- Administration:

  2025 Fall: Committee of Environmental Protection, Occupational Safety and Health, NTUCE

  2026 Spring: Committee of Environmental Protection, Occupational Safety and Health, NTUCE
  Committee of Preparation, 18NCSE & 8NCEE

University of California San Diego (UCSD), US

Instructor/ Lecturer (07/2023 - 08/2023)

- Course: Structural Analysis I

Teaching Assistant (Spring 2021 & Spring 2023)

- Course: Structural Reliability, Random Vibration

Graduate Research Assistant (09/2019 – 01/2025)

Validate the Design of the Upgrade Project for the World's Largest Outdoor Shake Table (LHPOST)

- Calculated the capability of the shake table system (acceleration, velocity, and displacement in 6-DOF space) based on the component specs provided by the MTS Corp.
- Developed a toolkit programmed in MATLAB to visualize the kinematics of the shake table system, so that users can check if a time history is achievable without collisions inside the system.

Assist the On-site Work during the Upgrade Project and the Acceptance Testing

- Documented UCSD (the experimental facility) and MTS Corp. (the system provider) works and weekly meeting during dissembles and upgrade.
- Defined the acceptance test plan with criteria and negotiated with MTS Corp. The test plan includes white noise tests, triangular wave tests, sinusoidal wave tests, and ground motion tests.
- Analyzed the recorded shake table data during the acceptance tests to help UCSD and NSF determine if the upgrading work passed the criteria.

Implement the UCSD Sensor Plan for the Shake Table System and the Characterization Testing

- Ordered the pressure transducers and hydraulic adapter to monitor the actuator hydraulic pressure and made the cables to connect the transducers to the MTS controller
- Defined the characterization test plan and identified the physical parameters of the system.

Develop a Comprehensive Numerical Modeling of the 6-DOF Shake Table System

- Twists and wrenches, an efficient way to represent 6-DOF kinematics and dynamics, is introduced to study the platen dynamics
- Various sources of nonlinearity are considered, including servo-hydraulic dynamics, ideal gas law for aero components, and nonlinear dissipative forces.

Study the Dynamic Interaction between the Shake Table System and the 10-story TallWood Building Specimen

- The coupling between two software program (the shake table dynamics in Matlab and the specimen dynamics in OpenSEES) is studied and a TCP/IP communication module is developed for the purpose.
- The specimen dynamics is studied by using OpenSEES (the Open System for Earthquake Engineering Simulation).

Apply and Develop Software for Laboratory Testing

- Identified the difference of data types between control system and data acquisition system such as MTS controller and NI instrumentation.
- Developed toolkits based on Python to convert decimal and binary data, system identification toolbox, and signal processors including time history shaping toolbox and triangular wave generator (indicated later in the software collections).

National Center for Research on Earthquake Engineering (NCREE), Taiwan

Junior Researcher/Assistant Technician (09/2016 - 08/2019)

Procure the 2m×2m Stewart-platform Type Shaking Table Testing System

- Evaluated the requirements for testing wind turbine parts and supporting structure, including stroke limit in 6 degree-of-freedom and acceleration capacity across the control bandwidth.
- Determined the target time history for acceptance tests, including random vibrations, sinusoidal motions and earthquake time histories, as well as the criteria for the performance testing.

Research on Experimental Technology and Hydraulic System

- Coordinated principal investigators of a joint research project on development and implementation of advanced experimental technology for testing of innovative anti-seismic components.
- Provided technical support of the small-scale testing facilities in NCREE Taipei Laboratory for new experimental technology validation and small-scale component testing.

Design Passive Transfer System for Shaking Table Testing

- Redesigned the proof-of-concept transfer system which could introduce additional vertical movement. Instead of using the friction-pendulum type, the new design is the spring-mass type.
- Designed the special-purpose springs for this system with long stroke and long original length. The design flow chart were also developed for users to design a similar transfer system.
- Improved the feedback controllers with adaptive control algorithms. The robustness of parameter identification and stability were considered.

Perform Shaking Table Testing and Laboratory Services

- 1st generation chief operator of the 8m×8m 6DOF Shaking Table Test System at NCREE Tainan Laboratory which was opened in 2017.
- Participated in many industrial and research projects conducting shaking table tests for seismic qualification of the nonstructural and substructure components based on the GR-63-CORE Codes or near-fault earthquakes.
- Conducted time history shaping, system identification, and project scheduling.

Department of Civil Engineering, National Taiwan University (NTU), Taiwan

Teaching Assistant, (Fall/2014)

Course: Structural Engineering and Fluid Mechanics Lab

Graduate Research Assistant (2014 - 2016)

Research on Floor Response Reproduction of High-rise Buildings

- Designed the proof-of-concept transfer system based on friction-pendulum system which was used to amplify the stroke limit of an existing shaking table. System identification was also performed to acquire the properties and parameters based on the linear and nonlinear models.
- Conducted the finite element model analysis of the rigid platform and the examination of the component capacity.
- Developed the linear controllers including feedforward, PID and LQG controls to verify the feasibility and controllability of the passive transfer system.
- Improved the system performance using the non-linear controllers including adaptive feedforward control, magnitude matching and ξ-scheduling methods.

Research on Shaking Table Performance Improvement and Real-time Control Method

- Checked the acceleration-tracking performance of an uni-axial shaking table with displacement control by using transfer function and RMS error. This system consists of a 50-ton actuator which was manufactured by local company using a servo valve made by MOOG Corporation.
- Improved the performance of the table by model-based acceleration feedback control. The model represented the relationship between the displacement input and acceleration output. The controllers were designed based on the feedforward and robust control algorithms.

Research on the Multi-roller Bearing (MRB)

- Implemented the active control method on the MRB. MRB was one kind of small-scale isolator for valuable goods or devices. The control force provided by the electromagnet was added to adjust the normal force on the friction surface.
- Simulated the performance tests of the multi-roller bearing after implementing the control force via MATLAB and Simulink. The dynamics of the system and the control force were also added to the simulation model based on a linear equation of motion.

Undergraduate Research Assistant, (Spring/2013)

Software Improvement of Brace on Demand

- Validated the design criteria and codes implemented in the BOD (Brace on Demand). BOD was developed by Tsai, Keh-Chyuan research group and was used to design buckling-restrained braces and connections given the defined axial load capacity, beam and column dimensions.
- Developed toolkits to find out computing error and identified reasons of program crashing.

AWARDS

Application of Active Control Technology to the Taipei 101 Wind Damper System for Displacement Suppression and Energy Dissipation Improvement **2026**

National Science and Technology Council, Taiwan

5-year Yushan (Young) Scholar Program **2025**

Ministry of Education, Taiwan

Funding from NEHRI@UCSD Large High Performance Outdoor Shake Table **2023 ~ 2024**

National Science Foundation Funding, US

Government Scholarships for Study Abroad **2021 ~ 2022**

Ministry of Education, Taiwan

Honorable Mention Award **2021**

2nd International Competition for Structural Health Monitoring

J. Yang Scholarship **2020 ~ 2021**

University of California San Diego, US

Excellent Presentation Award **7/6/2016**

7th Monitoring and Control in Civil Engineering, Taiwan

JOURNAL PUBLICATIONS

1. Lai, C. T., & Conte, J. P. (Under preparation). Verification of the Dynamic Model of the UC San Diego NHERI Six-Degree-Of-Freedom Large High-Performance Outdoor Shake Table under Loaded Table Condition.
2. Lai, C. T., & Conte, J. P. (2025). Calibration and Validation of a Dynamic Model of the UC San Diego NHERI 6-DOF Large High-Performance Outdoor Shake Table. *Engineering Structures*, Volume 343,

Part C, 2025, 121054.

3. McCartney, J. S., Van Den Einde, L., Morrison, M., Palermo, A., Lotfizadeh, K., Lai, C. T., & Conte, J. P. (2025). Multi-degree of freedom shake table testing of large-scale structural and geotechnical systems with NHERI-UCSD LHPOST6. *Frontiers in Built Environment*, *11*, 1573390.
4. Lai, C. T., & Conte, J. P. (2024). Dynamic model of the UC San Diego NHERI six-degree-of-freedom large high-performance outdoor shake table. *Earthquake Engineering & Structural Dynamics*, *53*(15), 4511-4540.
5. Lai, C. T. Yeh, S. W., Weng, P. W., Hsiao, F. P., Huang B. Y. (2022). Kinematics and Identification Study Applied to the shaking Table by using the actuator measurement: take MAST for example. *Structural Engineering, Vol. 37 No.3,* pp.81-104.
6. Chen, P. C., & Lai, C. T. (2022). Advanced control strategy for floor response replication of high-rise buildings subjected to earthquakes. *Journal of Earthquake Engineering*, *26*(3), 1228-1252.
7. Van Den Einde, L., Conte, J. P., Restrepo, J. I., Bustamante, R., Halvorson, M., Hutchinson, T. C., Lai, C. T. ... & Tsampras, G. (2021). NHERI@ UC San Diego 6-DOF large high-performance outdoor shake table facility. *Frontiers in Built Environment*, *6*, 580333.
8. Chen, P. C., Kek, M. K., Hu, Y. W., & Lai, C. T. (2018). Statistical reference values for control performance assessment of seismic shake table testing. *Earthquakes and Structures*, *15*(6), 595.
9. Chen, P. C., Lai, C. T., & Tsai, K. C. (2017). A control framework for uniaxial shaking tables considering tracking performance and system robustness. *Structural Control and Health Monitoring*, *24*(11), e2015.

TECHNICAL REPORTS

1. Lai, C. T., & Conte, J. P. (2024). *Dynamic modeling of the UC San Diego NHERI six-degree-of-freedom Large High-Performance Outdoor Shake Table*. PEER Report No. 2024/07, Pacific Earthquake Engineering Research Center, University of California, Berkeley, CA.
2. Lai, C. T., Huang, B. Y., Yeh, S. W., … & Cai, J. F. (2021). Identification Study of 6DOF Shaking Table Systems (MAST). *NCREE-2021-002.*
3. Lu, L. Y., Hsiao, K. A., Wang, S. J., …Lai, C. T., … & Lin, G. L. (2020). Dynamic Characteristics Identification and Performance Assessment of Floating Foundation of Shaking Table in NCREE Tainan Laboratory. *NCREE-2020-003.*
4. Weng Y. T., Hsiao, F. P., Chou, C. C., … Lai, C. T., … & Shen, W. C. (2019). Shaking Table Tests, Instrumental Technology, Structural Identification and Nonlinear Dynamic Analysis of a 3-story RC Frame Specimen. *NCREE-2019-004.*
5. Lai, C. T., Lin, C. C., Chen, P. C., Yang, H. H. (2017). Manual of Uni-axial Shaking Table Simulator (469USC). *NCREE-2017-002.*
6. Lai, C. T., Chen, P. C., Tsai, K. C. (2016). System Control Strategy on Floor Response Reproduction of High-rise Buildings Subjected to Earthquakes. *NCREE-2016-011.*
7. Lai, C. T., Chen, P. C., Lin, P. Y., Chiang, H. W., Hwang, J. S. (2015). A Study on Improving the Acceleration Tracking Performance of a Uni-axial Shaking Table by Employing Control Theories. *NCREE-2015-018.*

CONFERENCE PUBLICATIONS (* denotes speaker)

1. Lai, C. T.* & Conte, J. P. (2026). Numerical Investigation of Control–Table–Specimen Dynamic Interaction for NHERI-UCSD 6-DOF Large High-Performance Outdoor Shake Table. *Engineering Mechanics Institute Conference 2026. June 2-5, Boulder, Colorado.*
2. Lai, C. T.* & Conte, J. P. (2025). NHERI−UCSD Large High-Performance Outdoor Shake Table – Digital Twin, Calibration, and Control-Structure Interaction. *ACMT Symposium on Machine Learning and Digital Twin Technologies Against Natural Disasters. Dec 12-13, Tainan, Taiwan.*
3. Lai, C. T.* & Conte, J. P. (2025). NHERI−UCSD 6-DOF Large High-Performance Outdoor Shake

Table – Upgrade, Digital Twin, and Tall Building Tests. *Fifth International Conference on Sustainable Infrastructure and Built Environment. Nov 4-6, Bandung, Indonesia.*

4. Lai, C. T.* & Conte, J. P. (2025). An Iteration Strategy Using Fully Numerical Simulation for Improving Acceleration Tracking Performance of Shake Table Systems. *Engineering Mechanics Institute Conference 2025. May 27-30, Anaheim, California.*

5. Lai, C. T.* & Conte, J. P. (2024). Modeling of the NHERI UCSD 6-DOF Outdoor Shake Table under Bare Table and Loaded Conditions. *18th World Conference on Earthquake Engineering (WCEE2024). June 30-July 5, Milan, Italy.*

6. Lai, C. T.* & Conte, J. P. (2024). Validation of LHPOST6 Shake Table-Specimen Model using Shake Table Tests data. *Engineering Mechanics Institute Conference 2024. May 28-31, Chicago,, United States.*

7. Lai, C. T.* & Conte, J. P. (2023). Modeling of the Dynamic Interaction between the NHERI−UCSD 6-DOF Large High-Performance Outdoor Shake Table and TallWood Building Specimen. *Engineering Mechanics Institute Conference 2023. June 6-9, Atlanta, United States.*

8. Lai, C. T.* & Conte, J. P. (2022). Modeling of NHERI-UCSD Upgraded 6-DOF Large High-Performance Outdoor Shake Table. *Engineering Mechanics Institute Conference 2022. May 31-June 3, Baltimore, United States.*

9. Lai, C. T.*, Chen, P. C. (2018). Methods of Reproducing Floor Response of High-rise Buildings subjected to Ground Motions. *The 14th National Conference on Structural Engineering and the 14th National Conference on Earthquake Engineering, Taichung, Taiwan*.

10. Chen, P. C., Lai, C. T.* & Tsai, K. C. (2017). Development of Advanced Experimental Technology on Reproducing Floor Response of High-rise Buildings subjected to Ground Motions. *7th International Conference on Advances in Experimental Structural Engineering. September 6-8, 2017, Pavia, Italy.*

11. Lai, C. T.*, Chen, P. C. & Tsai, K. C. (2016). Shaking Table Control Strategy on Floor Response Reproduction of High-rise Buildings Subjected to Earthquakes. *The 29th KKHTCNN Symposium on Civil Engineering, December 3-5, 2016, Hong Kong, China*.

12. Lai, C. T.*, Chen, P. C. & Tsai, K. C. (2015). Improved Tracking Performance of a Uni-axial Seismic Shaking Table by Introducing Control Theory, Proceedings, *The 28th KKHTCNN Symposium on Civil Engineering, November 16-18, 2015, Bangkok, Thailand*.

13. Chen, P. C.*, & Lai, C. T. (2015). Alternative Acceleration Performance Assessment Method for Seismic Shaking Tables. *6th International Conference on Advances in Experimental Structural Engineering, August 1-2, 2015, University of Illinois, Urbana-Champaign, United States.*

14. Chen, P. C.*, Wang, S. J., Shiau, C. Y., & Lai, C. T. (2014). Smart Rolling-type Isolation Devices Using Embedded Electromagnet. *The 5th Asia Conference on Earthquake Engineering, October 16-18, 2014.*

SEMINAR SPEECH

1. The past, present, and future: NHERI UCSD Large High Performance Outdoor Shake Table (2026). *INTI International University, July 11, 2026, Negeri Sembilan, Malaysia*

2. 大型振動台試驗的前世與今生 (2025). 國立中興大學土木工程學系，民國一一四年十一月二十日，臺中臺灣。

INTERNATIONAL SITE VISITS

1. The 10th Asian Civil Engineering Conference (CECAR10), Oct 21-24, 2025 Jiju, South Korea

2. Seismic Research and Test Center (SESTEC), July 6-8, 2026, Pusan, South Korea

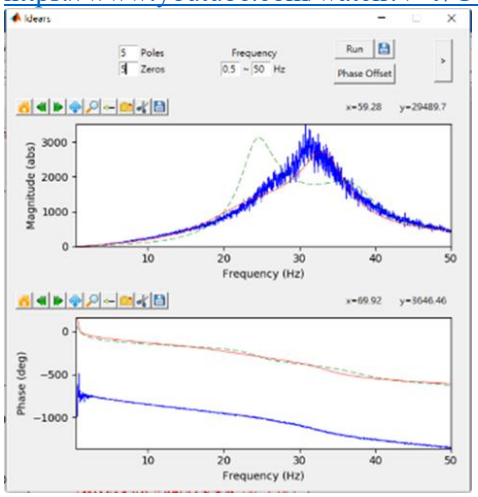
3. The 19th East Asia-Pacific Conference on Structural Engineering and Construction (EASEC-19), July 13-15, 2026, Kuala Lumpa, Malaysia

TECHINICAL SKILLS

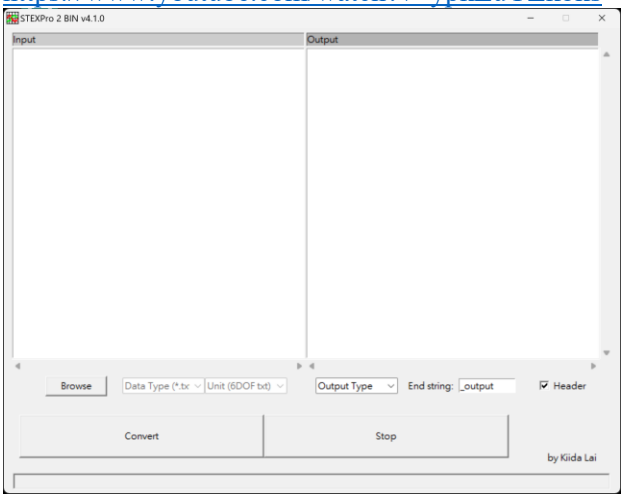
1. Programming languages and analysis tools: Matlab, C/C++, Python.
2. Control software: MTS 469D, STEXPro, MOOG Test Suite, dSPACE DS1103, MicroLabBox.
3. Computer aided design/engineering: ETABS, PISA3D, Abaqus, AutoCAD, SketchUp.
4. Research area: Structural control, Robotics, System identification, Experimental technology.

PORTFOLIO

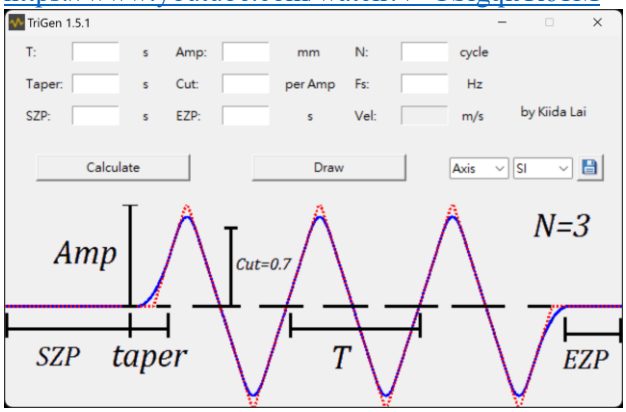
dSPACEinPython
https://www.youtube.com/watch?v=t7U8WVCQe28



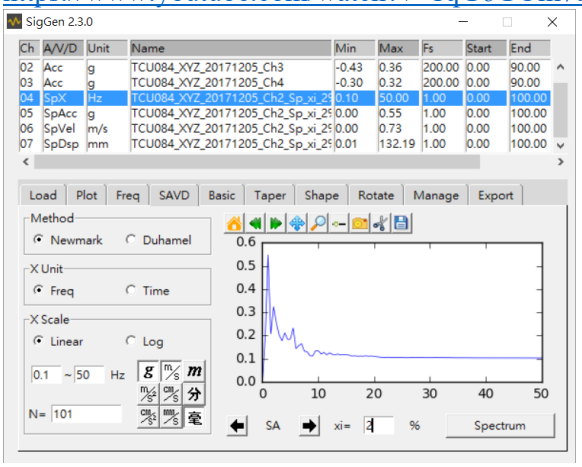
STEX2BIN
https://www.youtube.com/watch?v=ypkZuYZkbks



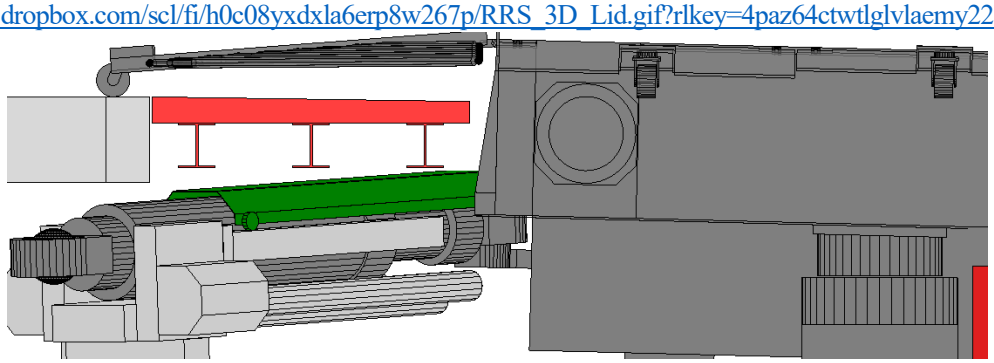
TriGen
https://www.youtube.com/watch?v=CSigqiJR61M



SigGen
https://www.youtube.com/watch?v=1qG6GOm7t6g



Shake table kinematics in Matlab
https://www.dropbox.com/scl/fi/h0c08yxdxla6erp8w267p/RRS_3D_Lid.gif?rlkey=4paz64ctwtlglvlaemy22hqzj&dl=0



Appendix: Student Evaluation of UCSD SE130A – *Structural Analysis I* in Summer 2023

Questions		Score: 5	4	3	2	1
1.	How often did you attend class and/or engage with course materials?	5	5	1	0	0
2.	Class sessions helped me understand the course content.	7	4	0	0	0
3.	Assignments helped me understand the course content.	5	6	0	0	0
4.	Exams and other assessments were a fair measure of course learning outcomes.	5	5	1	0	0
5.	I received timely and helpful feedback in this course.	5	6	0	0	0
6.	The course developed my understanding and skills for the subject.	5	6	0	0	0
7.	The course was intellectually stimulating and engaging.	4	6	1	0	0
8.	The instructor effectively communicated course goals, learning outcomes, and expectations.	6	5	0	0	0
9.	There were sufficient opportunities for me to get help when needed via offered opportunities.	7	4	0	0	0
10.	The instructor incorporated teaching methods that helped me learn.	5	6	0	0	0
11.	The instructor arrived on time to class sessions and office hours.	6	5	0	0	0
12.	The instructor emphasized the importance of academic integrity.	7	4	0	0	0

13. Please describe any specific aspects of the course and/or teaching practices that your instructor used that particularly helped you to learn the material and/or develop your own critical perspectives on the material.

- Instructor was not focused on conceptual work. While I do enjoy conceptual side and it helps me learn, understanding why the equations are the way it is, I believe his strategy of teaching through examples, especially in these quick summer sessions, was very helpful.
- Professor Lai has the best notes in all of the course i have ever attended at ucsd.
- The instructor is always willing to help when needed.
- The outline notes for class were super helpful. I know you said they are very time consuming to prepare so I'm very appreciative of your effort. Also the ways you broke apart the solution derivation into like 1-6 straight forward steps made problem solving much less daunting.
- The professor helped me with my assignments and understanding the concepts by consistently uploading the lectures notes and previous homework solutions. This allowed me to look back at what I might have missed and further practice the problems.

14. Please describe any specific aspects of the course and/or teaching practices that your instructor used that helped you to feel engaged with the course material or that encouraged you to feel that you could succeed in the course.

- During lectures, the instructor allowed time for questions on new topics/methods introduced. He also took time to address the exam questions after taking the tests and gave us hints on some difficult questions on the homework.
- he answers questions very well and has perfect notes that anyone can follow
- Need to do more harder examples,
- The professor gave good examples through the problem sets and exams were on par with what was done in class. Being detailed in that and offering the time to help was good.
- You always replied really quick to piazza posts and emails. I'm sure it's annoying to be so on top of responses but I appreciated that a lot.

15. Please describe any specific aspects of the course and/or teaching practices that your instructor used that were less helpful for your learning. Optionally you may offer constructive suggestions that might improve their effectiveness.

- i wish he can teach more se courses
- My only issue was just doing examples and not tying it to concepts makes it harder to understand the topic. I think the professor did a good job teaching this class, especially for a summer course. I'm not sure there is much else he could improve besides that.
- Thank you so much for your teaching

16. Is there anything else you would like to share about your experience in the course?

- I really liked the outline notes and I think they were a big reason I'm able to feel confident in my understanding of the course content.
- You should become a professor, you teach very well especially given how hard the course is

17. Hours per week of work outside of class.

- 1-5 hours: 2 students
- 6-10 hours: 6 students
- More than 10 hours: 3 students

18. Expected grade in the course.

- B: 8 students
- C: 3 students